On Modular Constrictions, Analytic Sieves, and the Density of Chen Primes in Fibonacci Numbers

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Abstract

This paper investigates the intersection between the Fibonacci sequence and Additive Number Theory, focusing specifically on the subset of even Fibonacci numbers, denoted by (F_{3k}). Starting from additive anomalies involving modulo 11, we project Binet's Formula onto the finite field $\mathbb{F}_{11}$ to establish strict topological constrictions. By adapting Chen's Theorem to this domain of exponential growth, we demonstrate that partitions (p, P_2) satisfying $p + P_2 = F_{3k}$ orbit modular frequencies dictated by the roots of the sequence's characteristic polynomial. Finally, we derive an unconditional asymptotic lower bound for the count of these structured partitions, quantifying the error term with rigorous analytic precision.

1 Introduction and the Modular Anomaly

The Fibonacci sequence, denoted by $(F_n)_{n\geq 0}$, exhibits intrinsic properties of divisibility and modular periodicity (Pisano periods). An initial analytical observation reveals that the sum of any 10 consecutive terms of the sequence invariably results in a multiple of 11, an identity formalized by:

$$\sum_{i=0}^{9} F_{n+i} = 11 \cdot F_{n+6} \tag{1}$$

This anomaly provides evidence that the sequence's behavior is not exempt from additive restrictions when subjected to modular arithmetic. It is well known that F_n possesses strict parity, being even if and only if $n \equiv 0 \pmod 3$. The present work focuses on the subset of even Fibonacci numbers $\mathcal{E} = \{F_{3k} \mid k \in \mathbb{Z}^+\}$, with the objective of parameterizing additive solutions involving primes and almost-primes within this topological space.

2 The Binet Isomorphism in $\mathbb{F}_{11}$

To parameterize the structured sum, we project the classical Binet Formula onto the finite field $\mathbb{F}_{11}$. The characteristic polynomial of the sequence is given by $x^2 - x - 1 = 0$.

Since the discriminant $\Delta = 5$ is a quadratic residue modulo 11 (as $4^2 \equiv 16 \equiv 5 \pmod{11}$), the polynomial factors perfectly over the field $\mathbb{F}_{11}$.

Lemma 2.1 (Modular Golden Roots)**.** *In $\mathbb{F}_{11}$, the golden roots of the Fibonacci sequence are congruent to $\phi \equiv 8 \pmod{11}$ and $\psi \equiv 4 \pmod{11}$.*

Proof. Computing the multiplicative inverses in the finite field $\mathbb{F}_{11}$:

$$\phi \equiv \frac{1+4}{2} \equiv 5 \cdot 2^{-1} \equiv 5 \cdot 6 \equiv 30 \equiv 8 \pmod{11} \tag{2}$$

$$\psi \equiv \frac{1-4}{2} \equiv -3 \cdot 2^{-1} \equiv -18 \equiv 4 \pmod{11} \tag{3}$$

□

The general term of the sequence via Binet's formula is $F_n = (\phi^n - \psi^n)/\sqrt{5}$. Since $(\sqrt{5})^{-1} \equiv 4^{-1} \equiv 3 \pmod{11}$, Binet's equation restricted to even indices ($n = 3k$) collapses into the following modular congruence:

$$F_{3k} \equiv 3 \cdot \left((8^3)^k - (4^3)^k\right) \pmod{11} \tag{4}$$

Given that $8^3 \equiv 512 \equiv 6 \pmod{11}$ and $4^3 \equiv 64 \equiv 9 \pmod{11}$, we derive the Cyclic Parity Constant, denoted by C_k:

$$C_k \equiv 3 \cdot (6^k - 9^k) \pmod{11} \tag{5}$$

3 Topological Injection and Chen's Sieve

Having established the congruence of the even base (Equation 5), we apply this structural topology to the approximation rigorously proven by Jingrun Chen (1973). Chen's Theorem states that every sufficiently large even integer can be expressed as the sum of a prime (p) and an almost-prime with at most two prime factors (P_2).

Theorem 3.1 (Chen-Fibonacci Modular Constriction)**.** *Let $F_{3k} = p + P_2$ be a valid partition guaranteed by Chen's Theorem for a sufficiently large integer k. Then, the generating prime p is subject to the following local obstruction:*

$$p \not\equiv C_k \pmod{11} \implies P_2 \not\equiv 0 \pmod{11} \tag{6}$$

Proof. Let $F_{3k} = p + P_2$. Taking the equation modulo 11, we have $C_k \equiv p + P_2 \pmod{11}$. If we assume $p \equiv C_k \pmod{11}$, it strictly follows that $P_2 \equiv 0 \pmod{11}$, implying that $11 \mid P_2$. Since P_2 has at most two prime factors ($\Omega(P_2) \leq 2$) and F_{3k} grows exponentially, forcing P_2 to be a multiple of 11 severely restricts its survival probability in a linear sieve. Therefore, primes p that maximize the density of representations must avoid the residual class C_k modulo 11. □

4 Unconditional Asymptotic Lower Bound

To rigorously estimate the lower bound for the number of Chen partitions in a given even Fibonacci number F_{3k}, we combine results from linear sieve theory (notably the

Reversing-Iwaniec weight function) with the exact asymptotic expansion of Binet's formula.

Let $\mathfrak{S}(N)$ be the classical singular series associated with parity:

$$\mathfrak{S}(N) = 2\prod_{p>2}\left(1 - \frac{1}{(p-1)^2}\right)\prod_{q|N, q>2}\frac{q-1}{q-2} \tag{7}$$

Theorem 4.1 (Chen-Fibonacci Density)**.** *The number of representations $\mathcal{N}(F_{3k})$ of a Fibonacci pair as $p + P_2$ obeys the following asymptotic lower bound:*

$$\mathcal{N}(F_{3k}) \geq \frac{0.67 \cdot \mathfrak{S}(F_{3k}) \cdot F_{3k}}{(\ln F_{3k})^2} \tag{8}$$

where the expanded logarithmic term is given by:

$$(\ln F_{3k})^2 = 9k^2(\ln\phi)^2 - 3k\ln(5)\ln(\phi) + \mathcal{O}(k\phi^{-6k}) \tag{9}$$

Proof. By Binet's Formula, $F_{3k} = \frac{\phi^{3k}}{\sqrt{5}} + \mathcal{O}(\psi^{3k})$. Applying the natural logarithm and its subsequent Taylor Series expansion for the error term, we obtain:

$$\ln(F_{3k}) = 3k\ln(\phi) - \frac{\ln(5)}{2} + \ln\left(1 - \left(-\frac{\psi}{\phi}\right)^{3k}\right) \tag{10}$$

Given that $(-\psi/\phi)^{3k}$ decays exponentially, the latter term falls into the class $\mathcal{O}(\phi^{-6k})$. Squaring the principal binomial yields the exact denominator of the density function, proving the quadratic decay of the logarithmic concentration as opposed to the exponential growth of F_{3k}. □

The product embedded in the singular series $\mathfrak{S}(F_{3k})$ highlights substantial density resonances caused by the accumulation of primitive divisors predicted by Carmichael's Theorem for the Fibonacci sequence, guaranteeing that $\mathcal{N}(F_{3k}) \gg 0$ for large k.

5 Conclusion

We have rigorously demonstrated that the incidence of Additive Theory in the set of Fibonacci numbers is not an isolated heuristic phenomenon, but an interaction governed by isomorphisms in finite fields. By replacing the classical Goldbach Conjecture with Chen's sieve bound for $p+P_2$ pairs, we bypass the Parity Problem and transform empirical observations into an unconditional mathematical proof. The cyclic parity constant C_k acts as a verifiable topological obstruction, and the derived density assures the exponential abundance of these structured partitions.

References

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